# Chapter 3 — Indefinite Integration: Last-Minute Revision

Maharashtra HSC · Maths Part II · Exam-tomorrow notes · File 2 of 3

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## 0. Read This First

**Assumption:** your exam is tomorrow and you remember little. This file is self-contained: each block gives the idea in two lines, the formula, and three worked examples (Easy → Medium → Hard) for every problem type in the chapter. Total reading time ≈ 75 minutes.

**What the board does (verified from your six papers, Mar 2024 – Jun 2026):** about 10 marks of Indefinite Integration questions in every paper — typically one 1-mark, one 2-mark, one 3-mark and one 4-mark question. 4-markers came from: partial fractions, $x^2 \sin 3x$ (by parts), the $\frac{1}{a\sin^2 x + b\cos^2 x + c}$ type, the $e^x[f+f']$ type, and proofs of standard formulas.

**Golden rule:** after every answer, *differentiate it in your head*. If you get the integrand back, you are right. Always write $+c$.

# 1. The Formula Sheet (memorise this page)

**Standard integrals**

| $f(x)$ | $\int f\,dx$ | $f(x)$ | $\int f\,dx$ |
|---|---|---|---|
| $x^n\ (n \neq -1)$ | $\frac{x^{n+1}}{n+1}$ | $\frac{1}{x}$ | $\log\lvert x\rvert$ |
| $e^x$ | $e^x$ | $a^x$ | $\frac{a^x}{\log a}$ |
| $\sin x$ | $-\cos x$ | $\cos x$ | $\sin x$ |
| $\sec^2 x$ | $\tan x$ | $\operatorname{cosec}^2 x$ | $-\cot x$ |
| $\sec x \tan x$ | $\sec x$ | $\operatorname{cosec} x \cot x$ | $-\operatorname{cosec} x$ |
| $\tan x$ | $\log\lvert\sec x\rvert$ | $\cot x$ | $\log\lvert\sin x\rvert$ |
| $\sec x$ | $\log\lvert\sec x + \tan x\rvert$ | $\operatorname{cosec} x$ | $\log\lvert\operatorname{cosec} x - \cot x\rvert$ |

**Linear rule:** $\int f(ax+b)\,dx = \frac{1}{a}\,g(ax+b) + c$. **Never forget the $\frac{1}{a}$.**

**Special seven (proofs of 3 and 4 have been asked)**

| Integrand | Result | Integrand | Result |
|---|---|---|---|
| $\frac{1}{x^2+a^2}$ | $\frac{1}{a}\tan^{-1}\frac{x}{a}$ | $\frac{1}{\sqrt{a^2-x^2}}$ | $\sin^{-1}\frac{x}{a}$ |
| $\frac{1}{x^2-a^2}$ | $\frac{1}{2a}\log\left\lvert\frac{x-a}{x+a}\right\rvert$ | $\frac{1}{\sqrt{x^2-a^2}}$ | $\log\left\lvert x+\sqrt{x^2-a^2}\right\rvert$ |
| $\frac{1}{a^2-x^2}$ | $\frac{1}{2a}\log\left\lvert\frac{a+x}{a-x}\right\rvert$ | $\frac{1}{\sqrt{x^2+a^2}}$ | $\log\left\lvert x+\sqrt{x^2+a^2}\right\rvert$ |
| $\frac{1}{x\sqrt{x^2-a^2}}$ | $\frac{1}{a}\sec^{-1}\frac{x}{a}$ | | |

**Parts, roots and the shortcuts**

- **By parts:** $\int u\,v\,dx = u\int v\,dx - \int\left(u'\int v\,dx\right)dx$; choose $u$ by **LIATE** (Log, Inverse trig, Algebraic, Trig, Exponential).
- $\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + c$
- $\int \sqrt{x^2\pm a^2}\,dx = \frac{x}{2}\sqrt{x^2\pm a^2} \pm \frac{a^2}{2}\log\left\lvert x+\sqrt{x^2\pm a^2}\right\rvert + c$
- $\int e^x[f(x)+f'(x)]\,dx = e^x f(x) + c$
- $\int e^{ax}\sin bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\sin bx - b\cos bx) + c$; $\int e^{ax}\cos bx\,dx = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx) + c$
- **Substitution corollaries:** $\int \frac{f'}{f} = \log|f|$; $\int f^n f' = \frac{f^{n+1}}{n+1}$; $\int \frac{f'}{\sqrt{f}} = 2\sqrt{f}$.

# 2. Which Method? — The 10-Second Recognition Guide

| You see… | Do this |
|---|---|
| Sum of simple terms, $\tan^2$, $\sin^2$, $\cos^3$, products like $\sin A\cos B$, $\sqrt{1\pm\sin}$ | **Simplify** with identities (Type 1) |
| An inner function *and* its derivative ($x^9$ with $x^{10}$; $e^x(1+x)$ with $xe^x$; $\frac{1}{x}$ with $\log x$) | **Substitute** $t$ = inner function (Type 2) |

| You see… | Do this |
|---|---|
| Top = derivative of bottom | Answer = log\|bottom\| |
| $\frac{\sin(x+a)}{\cos(x-b)}$; $\frac{a\cos x+b\sin x}{c\cos x+d\sin x}$; powers of trig | **Trig types** (Type 3) |
| $\frac{1}{\text{quadratic}}$ or $\frac{1}{\sqrt{\text{quadratic}}}$ | **Complete the square** → special seven (Type 4) |
| $\frac{1}{a\sin^2 x+b\cos^2 x+c}$ | $\div\cos^2 x$, $t=\tan x$ (Type 5) |
| $\frac{1}{a\sin x+b\cos x+c}$ | $t=\tan\frac{x}{2}$ (Type 5) |
| $\frac{\text{linear}}{\text{quadratic}}$ or $\frac{\text{linear}}{\sqrt{\text{quadratic}}}$ | $N=A\cdot D'+B$ (Type 6) |
| Product of different families ($x\sin x$, $x\log x$, $e^x\cos x$), or a lone $\log x$, $\tan^{-1}x$ | **By parts** (Type 7) |
| $e^x\times$ (bracket) | Look for $f+f'$ (Type 8) |
| Polynomial ÷ factorisable polynomial | **Partial fractions** (Type 9) |

## 3. Type 1 — Simplify, Then Integrate

**Idea:** the integrand is a standard form in disguise. Expand, use identities, rationalise, or split the numerator using the denominator.

**Easy — $\int\frac{(x^2+1)^2}{x}\,dx$**

Expand: $\frac{x^4+2x^2+1}{x}=x^3+2x+\frac{1}{x}$.

$$\int\left(x^3+2x+\frac{1}{x}\right)dx=\frac{x^4}{4}+x^2+\log|x|+c.$$

**Medium — $\int\sin 4x\cos 3x\,dx$ (Ex 3.1 II.x)**

Product to sum: $2\sin 4x\cos 3x=\sin 7x+\sin x$.

$$I=\frac{1}{2}\int(\sin 7x+\sin x)\,dx=-\frac{\cos 7x}{14}-\frac{\cos x}{2}+c.$$

**Hard — $\int\frac{2}{\sqrt{x}-\sqrt{x+3}}\,dx$ (Ex 3.1 III.ix)**

Multiply by the conjugate $\sqrt{x}+\sqrt{x+3}$: the denominator becomes $x-(x+3)=-3$.

$$I=-\frac{2}{3}\int\left(x^{1/2}+(x+3)^{1/2}\right)dx=-\frac{2}{3}\left[\frac{2}{3}x^{3/2}+\frac{2}{3}(x+3)^{3/2}\right]+c=-\frac{4}{9}\left[x^{3/2}+(x+3)^{3/2}\right]+c.$$

**Board version [Feb 2026, 2 marks]: $\int\sqrt{1+\sin 2x}\,dx$**

$1+\sin 2x=(\sin x+\cos x)^2$, so the integral is $\int(\sin x+\cos x)\,dx=\sin x-\cos x+c$.

## 4. Type 2 — Substitution

**Idea:** find the inner function whose derivative is also present. Put it $=t$, convert $dx$, integrate, put $x$ back.

**Easy — $\int \frac{2x}{1+x^2}\,dx$ [PYQ Feb 2026, 1 mark]**

Top is the derivative of the bottom: $\log(1+x^2)+c$.

**Medium — $\int \frac{e^x(1+x)}{\sin^2(xe^x)}\,dx$ [PYQ Jul 2024, 2 marks]**

$\frac{d}{dx}(xe^x) = e^x(1+x)$. Put $xe^x = t$, $e^x(1+x)\,dx = dt$:

$$\int \operatorname{cosec}^2 t\,dt = -\cot t + c = -\cot(xe^x) + c.$$

($\cos^2$ version, Jun 2026: answer $\tan(xe^x)+c$.)

**Hard — $\int \frac{(x-1)^2}{(x^2+1)^2}\,dx$ (Ex 3.2 A I.14)**

Expand the numerator: $(x-1)^2 = (x^2+1) - 2x$. Split:

$$I = \int \frac{dx}{x^2+1} - \int \frac{2x}{(x^2+1)^2}dx.$$

Second part: $x^2+1 = t$, $2x\,dx = dt$: $\int t^{-2}dt = -\frac{1}{t}$.

$$I = \tan^{-1} x + \frac{1}{x^2+1} + c.$$

**Also asked**

- $\int x^9 \sec^2(x^{10})\,dx = \frac{1}{10}\tan(x^{10}) + c$ [Feb 2025]
- $\int \frac{dx}{x+x^{-10}} = \int \frac{x^{10}dx}{x^{11}+1} = \frac{1}{11}\log|x^{11}+1| + c$ [Jul 2025]
- $\int x^x(1+\log x)\,dx = x^x + c$ [Feb 2025 MCQ] (since $\frac{d}{dx}x^x = x^x(1+\log x)$)
- $\int \frac{dx}{\sqrt{x}} = 2\sqrt{x} + c$ [Jul 2025, 1 mark]

## 5. Type 3 — Trigonometric Types

**Ideas:** (i) rewrite the numerator's angle in terms of the denominator's angle; (ii) for $\frac{a\cos x + b\sin x}{c\cos x + d\sin x}$, write $N = \lambda D + \mu D'$; (iii) for powers, peel off one factor.

**Easy — $\int \frac{\sin x}{\cos^2 x}\,dx$**

$\frac{\sin x}{\cos^2 x} = \sec x \tan x$, so $I = \sec x + c$.

**Medium — $\int \frac{\sin(x+a)}{\cos(x-b)}\,dx$ [PYQ Jul 2024 and Jul 2025, 3 marks]**

$x+a = (x-b)+(a+b)$, so $\sin(x+a) = \sin(x-b)\cos(a+b) + \cos(x-b)\sin(a+b)$. Divide by $\cos(x-b)$:

$$I = \int [\cos(a+b)\tan(x-b) + \sin(a+b)]\,dx = \cos(a+b)\log|\sec(x-b)| + x\sin(a+b) + c.$$

**Hard — $\int \frac{2\cos x + 3\sin x}{4\cos x + 5\sin x}\,dx$**

$D = 4\cos x + 5\sin x$, $D' = 5\cos x - 4\sin x$. Write $2\cos x + 3\sin x = \lambda D + \mu D'$:

$$\cos x:\ 4\lambda + 5\mu = 2, \qquad \sin x:\ 5\lambda - 4\mu = 3.$$

$\Rightarrow 41\lambda = 23, \lambda = \frac{23}{41}, \mu = \frac{2-4\lambda}{5} = -\frac{2}{41}$.

$$I = \lambda x + \mu \log|D| + c = \frac{23}{41}x - \frac{2}{41}\log|4\cos x + 5\sin x| + c.$$

**Powers in one line each**

$\cos^3 x = \frac{1}{4}(\cos 3x + 3\cos x)$; $\sin^2 x = \frac{1-\cos 2x}{2}$; $\tan^2 x = \sec^2 x - 1$; odd power of cos: keep one $\cos x$, convert the rest to sin, put $\sin x = t$.

## 6. Type 4 — Special Integrals and Completing the Square

**Idea:** make the $x^2$ coefficient 1, add and subtract $\left(\frac{1}{2}\text{coef of } x\right)^2$, then use the special seven with $x \to (x+\beta)$.

**Easy — $\int \frac{dx}{25-9x^2}$ [PYQ Feb 2025, 2 marks]**

$$\frac{1}{9}\int \frac{dx}{\left(\frac{5}{3}\right)^2 - x^2} = \frac{1}{9}\cdot\frac{1}{2\cdot\frac{5}{3}}\log\left|\frac{\frac{5}{3}+x}{\frac{5}{3}-x}\right| + c = \frac{1}{30}\log\left|\frac{5+3x}{5-3x}\right| + c.$$

**Medium — $\int \frac{dx}{\sqrt{5-4x-x^2}}$**

$5 - 4x - x^2 = 5 - (x^2+4x+4) + 4 = 9 - (x+2)^2$.

$$I = \int \frac{dx}{\sqrt{3^2 - (x+2)^2}} = \sin^{-1}\frac{x+2}{3} + c.$$

**Hard — $\int \frac{dx}{x[(\log x)^2+4\log x-1]}$**

Put $\log x = t$, $\frac{dx}{x} = dt$: $\int \frac{dt}{t^2+4t-1} = \int \frac{dt}{(t+2)^2-(\sqrt{5})^2}$.

$$I = \frac{1}{2\sqrt{5}}\log\left|\frac{\log x + 2 - \sqrt{5}}{\log x + 2 + \sqrt{5}}\right| + c.$$

**Proof you may be asked [PYQ Mar 2024]: $\int \frac{dx}{a^2-x^2} = \frac{1}{2a}\log\left|\frac{a+x}{a-x}\right| + c$**

$\frac{1}{(a-x)(a+x)} = \frac{1}{2a}\left[\frac{1}{a-x} + \frac{1}{a+x}\right]$ (put $x = a$ and $x = -a$ in $1 = A(a+x) + B(a-x)$ to get $A = B = \frac{1}{2a}$).

$$I = \frac{1}{2a}\left[-\log|a-x| + \log|a+x|\right] + c = \frac{1}{2a}\log\left|\frac{a+x}{a-x}\right| + c.$$

## 7. Type 5 — Trigonometric Denominators

**Idea A** ($\sin^2$, $\cos^2$ in the denominator): divide by $\cos^2 x$, change $\sec^2 x \to 1 + \tan^2 x$, put $\tan x = t$. **Idea B** ($\sin x$, $\cos x$ to power 1 with a constant): $t = \tan\frac{x}{2}$, $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$, $dx = \frac{2dt}{1+t^2}$.

**Easy — $\int \frac{dx}{1+\cos^2 x}$**

Divide by $\cos^2 x$: $\int \frac{\sec^2 x\,dx}{\sec^2 x + 1} = \int \frac{\sec^2 x\,dx}{\tan^2 x + 2}$. Put $\tan x = t$:

$$\int \frac{dt}{t^2 + (\sqrt{2})^2} = \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{\tan x}{\sqrt{2}}\right) + c.$$

**Medium — $\int \frac{dx}{3+2\sin x}$ (Ex 3.2 B II.1)**

$t = \tan\frac{x}{2}$: denominator $\times(1+t^2)$ is $3(1+t^2)+4t$.

$$I = \int \frac{2\,dt}{3t^2+4t+3} = \frac{2}{3}\int \frac{dt}{\left(t+\frac{2}{3}\right)^2+\frac{5}{9}} = \frac{2}{3}\cdot\frac{3}{\sqrt{5}}\tan^{-1}\frac{t+\frac{2}{3}}{\frac{\sqrt{5}}{3}}+c = \frac{2}{\sqrt{5}}\tan^{-1}\left(\frac{3\tan\frac{x}{2}+2}{\sqrt{5}}\right)+c.$$

**Hard — $\int \frac{2\,dx}{3+2\sin^2 x+5\cos^2 x}$ [PYQ Jun 2026, 4 marks]**

Divide by $\cos^2 x$ and use $\sec^2 x = 1+\tan^2 x$:

$$I = \int \frac{2\sec^2 x}{3(1+\tan^2 x)+2\tan^2 x+5}dx = \int \frac{2\sec^2 x}{5\tan^2 x+8}dx.$$

$\tan x = t$: $I = \frac{2}{5}\int \frac{dt}{t^2+\frac{8}{5}} = \frac{2}{5}\cdot\frac{\sqrt{5}}{2\sqrt{2}}\tan^{-1}\frac{\sqrt{5}\,t}{2\sqrt{2}}+c = \frac{1}{\sqrt{10}}\tan^{-1}\left(\frac{\sqrt{5}\tan x}{2\sqrt{2}}\right)+c.$

**Also asked [Feb 2026, 3 marks]: $\int \frac{\sin x}{\sin 3x}\,dx$**

$\sin 3x = \sin x(3-4\sin^2 x)$, so $I = \int \frac{dx}{3-4\sin^2 x}$. $\div\cos^2 x$: $\int \frac{\sec^2 x}{3-\tan^2 x}dx = \frac{1}{2\sqrt{3}}\log\left|\frac{\sqrt{3}+\tan x}{\sqrt{3}-\tan x}\right|+c$.

## 8. Type 6 — Linear over Quadratic

**Idea:** write Numerator $= A\cdot\frac{d}{dx}(\text{quadratic})+B$. The $A$ part gives log (or $2\sqrt{}$); the $B$ part is Type 4.

**Easy — $\int \frac{4x+6}{x^2+3x+5}\,dx$**

$4x+6 = 2(2x+3)$ = twice the derivative of the denominator. $I = 2\log(x^2+3x+5)+c$.

**Medium — $\int \frac{2x+1}{x^2+4x-5}\,dx$ (Ex 3.2 C I.2)**

$2x+1 = (2x+4)-3$, so

$$I = \log|x^2+4x-5| - 3\int \frac{dx}{(x+2)^2-3^2} = \log|x^2+4x-5| - \frac{3}{6}\log\left|\frac{x-1}{x+5}\right|+c$$
$$= \log|x^2+4x-5| - \frac{1}{2}\log\left|\frac{x-1}{x+5}\right|+c.$$

**Hard — $\int \sqrt{\frac{x-7}{x-9}}\,dx$ (Ex 3.2 C I.6)**

Multiply inside by $\frac{x-7}{x-7}$: integrand $= \frac{x-7}{\sqrt{x^2-16x+63}}$. Now $x-7 = \frac{1}{2}(2x-16)+1$:

$$I = \frac{1}{2}\int \frac{2x-16}{\sqrt{x^2-16x+63}}dx + \int \frac{dx}{\sqrt{(x-8)^2-1}}$$
$$= \sqrt{x^2-16x+63} + \log\left|x-8+\sqrt{x^2-16x+63}\right|+c.$$

## 9. Type 7 — Integration by Parts

**Idea:** $\int uv = u\int v - \int(u'\int v)$, $u$ chosen by LIATE. A single function ($\log x$, $\tan^{-1}x$) is written as "function $\times 1$". Repeat if a power of $x$ remains.

**Easy — $\int x \log x\, dx$ [PYQ Jun 2026, 2 marks]**

$u = \log x$ (L before A).

$$I = \log x \cdot \frac{x^2}{2} - \int \frac{1}{x} \cdot \frac{x^2}{2} dx = \frac{x^2}{2} \log x - \frac{x^2}{4} + c.$$

($\int \log x\, dx = x \log x - x + c$ was asked in Mar 2024.)

**Medium — $\int x^2 \sin 3x\, dx$ [PYQ Jul 2024, 4 marks]**

$u = x^2$:

$$I = -\frac{x^2 \cos 3x}{3} + \frac{2}{3} \int x \cos 3x\, dx, \qquad \int x \cos 3x\, dx = \frac{x \sin 3x}{3} + \frac{\cos 3x}{9}.$$

$$I = -\frac{x^2 \cos 3x}{3} + \frac{2x \sin 3x}{9} + \frac{2 \cos 3x}{27} + c.$$

**Check with the tabular rule:** $x^2 \left(-\frac{\cos 3x}{3}\right) - 2x \left(-\frac{\sin 3x}{9}\right) + 2 \left(\frac{\cos 3x}{27}\right)$ ✓.

**Hard — $\int e^{-x} \cos 2x\, dx$ (Ex 3.3 II.2) — cyclic**

Parts twice with $u = e^{-x}$ both times:

$$I = e^{-x} \frac{\sin 2x}{2} + \frac{1}{2} \int e^{-x} \sin 2x\, dx = \frac{e^{-x} \sin 2x}{2} + \frac{1}{2} \left[ -\frac{e^{-x} \cos 2x}{2} - \frac{1}{2} I \right].$$

$I + \frac{1}{4} I = \frac{e^{-x}}{4} (2 \sin 2x - \cos 2x) \Rightarrow I = \frac{e^{-x}}{5} (2 \sin 2x - \cos 2x) + c.$
(Formula check: $a = -1, b = 2$: $\frac{e^{-x}}{5} (-\cos 2x + 2 \sin 2x)$ ✓.)

**Proof asked [PYQ Jul 2025, 4 marks]: $\int \sqrt{a^2 - x^2}\, dx$**

By parts with $u = \sqrt{a^2 - x^2}$, $v = 1$: $I = x\sqrt{a^2 - x^2} + \int \frac{x^2}{\sqrt{a^2 - x^2}} dx$. Write $x^2 = a^2 - (a^2 - x^2)$: the last integral is $a^2 \sin^{-1} \frac{x}{a} - I$. So $2I = x\sqrt{a^2 - x^2} + a^2 \sin^{-1} \frac{x}{a}$, giving $I = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + c$.

## 10. Type 8 — $\int e^x [f(x) + f'(x)]\, dx = e^x f(x) + c$

**Idea:** split the bracket into a function and its derivative; the answer is $e^x \times$ (the non-derivative piece). Proof: $\frac{d}{dx}[e^x f] = e^x (f + f')$.

**Easy — $\int e^x \left(\frac{1}{x} - \frac{1}{x^2}\right) dx$ (Ex 3.3 III.3)**

$f = \frac{1}{x}$, $f' = -\frac{1}{x^2}$. $I = \frac{e^x}{x} + c$.

**Medium — $\int \frac{x e^x}{(x+1)^2} dx$ (Ex 3.3 III.4)**

$\frac{x}{(x+1)^2} = \frac{(x+1)-1}{(x+1)^2} = \frac{1}{x+1} - \frac{1}{(x+1)^2}$: $f = \frac{1}{x+1}$, $f' = -\frac{1}{(x+1)^2}$.

$$I = \frac{e^x}{x+1} + c.$$

**Hard — $\int e^{\sin^{-1} x} \left(\frac{x + \sqrt{1-x^2}}{\sqrt{1-x^2}}\right) dx$ [PYQ Feb 2025, 4 marks]**

Put $\sin^{-1} x = t$: $x = \sin t$, $\sqrt{1 - x^2} = \cos t$, $\frac{dx}{\sqrt{1-x^2}} = dt$.

$$I = \int e^t (\sin t + \cos t)\, dt = e^t \sin t + c = x\, e^{\sin^{-1} x} + c.$$

**MCQ twin [Jun 2026]:** $\int \frac{\sin^{-1}x}{\sqrt{1-x^2}} e^{\sin^{-1}x}dx = \int te^t\,dt = e^{\sin^{-1}x}(\sin^{-1}x - 1) + c.$

## 11. Type 9 — Partial Fractions

**Idea:** split a proper fraction into simple pieces. Forms: $\frac{A}{x-a}$ for each linear factor; $\frac{A}{x-a} + \frac{B}{(x-a)^2}$ for a repeated one; $\frac{Bx+C}{x^2+px+q}$ for an irreducible quadratic. Find constants by putting roots, then comparing coefficients. **If degree(top) $\geq$ degree(bottom), divide first.**

**Easy — $\int \frac{dx}{x^2-5x+6}$**

$\frac{1}{(x-2)(x-3)} = \frac{1}{x-3} - \frac{1}{x-2}$ (cover-up: at $x=3$, $\frac{1}{3-2}=1$; at $x=2$, $\frac{1}{2-3}=-1$).

$$I = \log|x-3| - \log|x-2| + c = \log\left|\frac{x-3}{x-2}\right| + c.$$

**Medium — $\int \frac{3x^2+4x-5}{(x^2-1)(x+2)}\,dx$ [PYQ Feb 2026, 4 marks]**

$3x^2+4x-5 = A(x+1)(x+2) + B(x-1)(x+2) + C(x-1)(x+1)$. $x=1$: $2=6A$, $A=\frac{1}{3}$. $x=-1$: $-6=-2B$, $B=3$. $x=-2$: $-1=3C$, $C=-\frac{1}{3}$.

$$I = \frac{1}{3}\log|x-1| + 3\log|x+1| - \frac{1}{3}\log|x+2| + c.$$

**Hard — $\int \frac{5e^x}{(e^x+1)(e^{2x}+9)}\,dx$ [PYQ Mar 2024, 4 marks]**

$e^x = t$: $\int \frac{5\,dt}{(t+1)(t^2+9)}$. Set $5 = A(t^2+9) + (Bt+C)(t+1)$. $t=-1$: $A=\frac{1}{2}$. $t^2$: $A+B=0 \Rightarrow B=-\frac{1}{2}$. Constant: $9A+C=5 \Rightarrow C=\frac{1}{2}$.

$$I = \frac{1}{2}\log(e^x+1) - \frac{1}{4}\log(e^{2x}+9) + \frac{1}{6}\tan^{-1}\frac{e^x}{3} + c.$$

**Repeated factor in one line**

$\frac{3x-2}{(x+1)^2(x+3)} = \frac{11/4}{x+1} - \frac{5/2}{(x+1)^2} - \frac{11/4}{x+3}$; integral $= \frac{11}{4}\log\left|\frac{x+1}{x+3}\right| + \frac{5}{2(x+1)} + c$. Remember $\int \frac{dx}{(x+1)^2} = -\frac{1}{x+1}$.

## 12. Frequently Confused Pairs

| These look alike… | …but differ |
|---|---|
| $\int \frac{dx}{x^2+a^2}$ vs $\int \frac{dx}{x^2-a^2}$ | $\tan^{-1}$ vs $\log$ |
| $\int \frac{dx}{x^2-a^2}$ vs $\int \frac{dx}{a^2-x^2}$ | $\log\left\|\frac{x-a}{x+a}\right\|$ vs $\log\left\|\frac{a+x}{a-x}\right\|$ (positive square goes first) |
| $\int \frac{dx}{\sqrt{a^2-x^2}}$ vs $\int \sqrt{a^2-x^2}\,dx$ | $\sin^{-1}\frac{x}{a}$ vs the parts formula $\frac{x}{2}\sqrt{} + \frac{a^2}{2}\sin^{-1}\frac{x}{a}$ |
| $\int \frac{x\,dx}{x^2+a^2}$ vs $\int \frac{dx}{x^2+a^2}$ | $\frac{1}{2}\log(x^2+a^2)$ vs $\frac{1}{a}\tan^{-1}\frac{x}{a}$ |
| $\int \frac{dx}{(x+1)^2}$ vs $\int \frac{dx}{x+1}$ | $-\frac{1}{x+1}$ vs $\log\|x+1\|$ |
| $\int \sec x\,dx$ vs $\int \sec^2 x\,dx$ | $\log\|\sec x + \tan x\|$ vs $\tan x$ |
| $t = \tan x$ vs $t = \tan\frac{x}{2}$ | for $\sin^2, \cos^2$ (or $\sin 2x, \cos 2x$) vs for $\sin x, \cos x$ |

## 13. Mistakes That Cost Marks — and Exam Precautions

**Top 8 mark-losers**

1. No $+c$.
2. Forgetting $\frac{1}{a}$ in $\int f(ax+b)$, or $\frac{1}{3}$ in $\int \frac{dx}{25-9x^2}$-type questions.
3. Sign in $\int \frac{dx}{a-x} = -\log|a-x|$ and in $\int \sin$, $\int \operatorname{cosec}^2$.
4. Leaving $t$ in the final answer (always resubstitute).
5. Wrong LIATE choice — the integral gets harder.
6. Partial fractions: constant numerator over a quadratic (must be $Bx+C$); missing $\frac{A}{x-a}$ with a repeated factor; not dividing an improper fraction.
7. Completing the square without first making the $x^2$ coefficient 1.
8. Cyclic parts: changing the choice of $u$ on the second application.

**Presentation precautions**

- Write the substitution line explicitly: "Put $x^{10} = t, \therefore 10x^9 dx = dt$". It carries method marks.
- In partial fractions, show the identity $N = A(\ldots) + B(\ldots)$ and each value of $x$ you substitute.
- For "prove" questions, start from the left side and end with the exact printed result.
- In MCQs, write the option letter **and** the answer (e.g. "(b) $e^{\sin^{-1}x}(\sin^{-1}x - 1) + c$") — the Jul 2024 paper explicitly required both.

## 14. Rapid-Revision Checklist

Tick each line only if you can do it *without notes*:

- ☐ Write all 20 standard results (Section 1) in under 3 minutes.
- ☐ Integrate $\cos^3 x$, $\sin^2 x$, $\tan^2 x$, $\sin A \cos B$.
- ☐ Spot $\frac{f'}{f}$ instantly and write $\log|f|$.
- ☐ Derive $\int \tan$, $\int \cot$, $\int \sec$, $\int \operatorname{cosec}$.
- ☐ Prove $\int \frac{dx}{a^2-x^2}$ and $\int \frac{dx}{x^2+a^2}$.
- ☐ Complete the square for $ax^2+bx+c$ with $a \neq 1$ and $a < 0$.
- ☐ Do $\frac{1}{a\sin^2 x + b\cos^2 x + c}$ ($\div\cos^2 x$) and $\frac{1}{a\sin x + b\cos x + c}$ ($t = \tan\frac{x}{2}$).
- ☐ Solve $\frac{\sin(x+a)}{\cos(x-b)}$ and a $\lambda D + \mu D'$ question.
- ☐ Split $\frac{px+q}{\text{quadratic}}$ as $A \cdot D' + B$.
- ☐ By parts: $x\log x$, $x^2\sin 3x$, $e^{ax}\sin bx$, and prove $\int \sqrt{a^2-x^2}\,dx$.
- ☐ Recognise $e^x[f + f']$.
- ☐ Partial fractions: all three forms, plus the $e^x = t$, $\cos\theta = t$ and $x^2 = m$ conversions.

**Coverage note:** every exercise type in Ex 3.1 – 3.4 and the Miscellaneous Exercise maps to one of Types 1–9 above. Rare types not given separate examples here (they are in File 1): $\sqrt{\tan x} + \sqrt{\cot x}$ (§6.7), $\int (px+q)\sqrt{ax^2+bx+c}\,dx$, and $\int \frac{dx}{x\sqrt{x^2-a^2}}$.